Nihal Singh — Personal Knowledge Base for Jarvis AI

Document type: Personal Profile for Retrieval-Augmented Generation (RAG) **Last updated:** 23 October 2026 **Purpose:** Provide a personal AI assistant with reliable context about my background, technical skills, projects, learning goals, lifestyle, food preferences, and communication style.

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1. Personal Profile

My name is Nihal Singh. I am a Java backend developer with approximately 1.5–2 years of professional experience at Tata Consultancy Services (TCS), according to my career discussions.

My primary technical background is Java, Spring Boot, Spring Cloud, microservices, REST APIs, databases, and AWS. I am expanding my knowledge of AI application development, particularly Spring AI and retrieval-augmented generation.

Personal details

- **Name:** Nihal Singh
- **Age:** 24
- **Location context:** New Town, Kolkata, India, previously shared while planning Durga Puja visits.
- **Height:** Approximately 180 cm, as shared in September 2026.
- **Weight:** Approximately 64 kg, as shared in September 2026.
- **Diet:** Vegetarian.
- **Fitness goal:** Clean weight gain.
- **Primary career goal:** Become a Full Stack Java AI Developer.

Personal measurements and routines are snapshots from earlier conversations, not guaranteed to be current.

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2. Career Goals and Professional Direction

My long-term goal is to become a Full Stack Java AI Developer. I want to build on my Java backend experience rather than abandon it and start from scratch.

My desired skill combination includes:

- Java and modern Java features.
- Spring Boot and Spring Security.
- Spring Cloud and microservices architecture.
- Spring AI and generative AI integration.

- REST APIs and database design.
- Frontend development, including React.
- AWS cloud services.
- Docker, Kubernetes, Jenkins, and CI/CD.
- AI-powered applications using embeddings, vector databases, and RAG.

I am also interested in understanding Forward Deployed Engineer roles and determining how my existing backend development experience could transfer to them.

Career development priorities

I want to strengthen my fundamentals and practical implementation ability. I do not want to learn technologies only at a theoretical level; I want to understand how to implement, debug, deploy, and explain them in interviews.

I am particularly interested in integrating AI into existing Spring Boot applications because this allows me to combine my backend expertise with a growing area of software development.

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3. Technical Skills and Knowledge

Java

My Java learning and interview preparation includes:

- Java 8, Java 17, and Java 21.
- Object-oriented programming.
- Inheritance, abstraction, interfaces, and polymorphism.
- Collections and streams.
- Exception handling.
- Local variables, instance variables, and default values.
- Java date and time APIs, including `LocalDate`, `LocalTime`, and `Instant`.
- Reflection.
- Concurrency and thread safety.
- `ThreadLocal`, asynchronous execution, and JVM fundamentals.
- Design patterns and clean code.

I prefer explanations that describe why a language feature works, not merely its syntax.

Spring Boot and backend development

My backend development interests and experience include:

- Spring Boot application development.
- RESTful APIs.
- Spring Security and authentication.
- Hibernate and Spring Data JPA.
- Transaction management with `@Transactional`.
- Asynchronous processing with `@Async`.
- Application events using `ApplicationEventPublisher`.
- Aspect-oriented programming and custom annotations.
- Exception handling and custom exceptions.
- Method-level authorization using annotations such as `@PostAuthorize`.
- Password-reset and OTP flows.
- Access-token and refresh-token security.

I want to understand framework behavior, including proxies, bean lifecycle, transaction propagation, dependency injection, and how security context information flows through an application.

Microservices and Spring Cloud

I have worked with or explored the following concepts:

- Spring Cloud Gateway.
- Eureka service discovery.
- Spring Cloud Config Server.
- Load balancing.
- OpenFeign.
- Redis.
- Circuit breakers.
- Retry mechanisms.
- Event-driven microservices.
- Gateway filters and route configuration.
- Rate limiting.
- Resilience patterns and fallback endpoints.

I want to strengthen my understanding of exact implementation steps and production use cases. The fact that I have explored a concept should not automatically be interpreted as proof that I implemented it independently in production.

AWS and DevOps

My AWS-related experience and learning include:

- Amazon S3.
- Amazon RDS.
- Amazon EC2.
- Amazon ECS and ECR.
- Amazon SQS and SNS.
- IAM roles and permissions.
- AWS Secrets Manager.
- Amazon EKS.

My DevOps learning includes Docker, Jenkins pipelines, Kubernetes manifests, container image deployment, and CI/CD.

I prefer practical commands, beginner-friendly YAML, explanations of configuration choices, and verification steps that I can execute myself.

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4. Professional Experience at TCS

My professional project discussions include work involving a supermarket-chain customer. The customer name should not be included in public-facing documents unless disclosure is permitted.

AWS Secrets Manager

I worked on replacing a hard-coded password with a secret retrieved from AWS Secrets Manager.

I also addressed application startup resilience by introducing fixed-interval retries when retrieving the secret. The intention was to handle temporary service unavailability rather than fail immediately during startup.

This retry mechanism should not be described as solving permanent authentication or authorization failures. Invalid permissions or an incorrect secret still require investigation.

Hibernate pagination optimization

I worked on optimizing Hibernate pagination to avoid fetching excessive database records caused by a join before applying the page limit.

The approach discussed involved using two queries so that the requested page of records could be retrieved without loading unnecessary joined rows first.

I want to describe this work in technically defensible terms. I should not claim a specific percentage improvement unless I have measured and verified it.

Java version migration

Java 8-to-Java 21 migration has been discussed as part of my project context. The precise scope of my personal implementation and ownership should be confirmed before I claim sole responsibility for the migration.

Resume accuracy

When helping me prepare a resume, LinkedIn profile, or interview response:

- Do not exaggerate my professional experience.
- Do not invent performance metrics.
- Distinguish personal implementation from team-level work.
- Distinguish conceptual understanding from hands-on experience.
- Do not claim production ownership without supporting evidence.
- Prefer clear explanations that I can defend during follow-up questions.

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5. EasyBuy Demo Project

EasyBuy is a demo application with a Spring Boot backend and a separately developed frontend.

Architecture and technologies

- Spring Boot backend.
- Frontend communicating with the backend through APIs.
- Amazon S3 for image storage.
- Amazon RDS for relational database storage.
- S3 presigned multipart-upload URLs.
- IAM roles for AWS permissions.

Image upload flow

The backend generates presigned URLs for multipart uploads to S3. The frontend uses these URLs to upload image parts directly to S3 rather than routing all image bytes through the backend.

The backend coordinates the multipart upload and completes the upload workflow.

This design reduces the need for the backend to proxy large image payloads.

Security and deployment concepts

Project discussions have included running a Dockerized backend on EC2, configuring environment secrets, and placing RDS in a private subnet within a VPC.

These details should be checked against the current deployed implementation before being used as claims about the live environment.

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6. Spring AI and My Personal Jarvis

I am learning Spring AI to integrate generative AI into Spring Boot applications. One of my goals is to build a personal assistant, which I refer to as Jarvis.

Spring AI topics explored

- `ChatClient`.
- Prompt construction and system messages.
- OpenAI model integration.
- Streaming responses using `Flux<String>`.
- Advisors.
- Token usage and completion-token limits.
- Safeguard mechanisms.
- Conversation memory.
- Embeddings.
- Vector stores.
- Retrieval-augmented generation.
- Document retrieval and context injection.

I have experimented with retrieving documents from a vector store and incorporating their text into an AI prompt.

RAG architecture

My intended RAG workflow is:

1. Store personal information as documents with meaningful metadata.
2. Generate embeddings for those documents.
3. Save the embeddings and document content in a vector database.
4. Convert a user question into an embedding.
5. Retrieve the most relevant documents.
6. Provide the retrieved context to the language model.

7. Generate an answer grounded in the retrieved information.

I want Jarvis to retrieve relevant information about my career when I ask career questions, my dietary preferences when I ask about food, and my technical background when I ask about coding or interview preparation.

The assistant should not dump my entire profile into every prompt. It should retrieve only the information relevant to the current question.

AI models and cost awareness

I have experimented with OpenAI models, including GPT-4o-mini and GPT-5-nano, and have investigated token usage, response latency, token limits, and API costs.

I also have a Gemini API key and have explored using local language models through Ollama.

I use a MacBook Air with an Apple M4 chip. I am interested in running local models without unnecessarily consuming system resources when the application is not being used.

I want to understand how model selection, prompt size, retrieved context, output limits, and caching can affect cost and performance.

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7. Diet and Food Preferences

My dietary discussions have focused on vegetarian food and increasing protein intake to support clean weight gain.

Dietary preferences

- Vegetarian diet.
- Prefer increasing protein without relying on paneer or tofu.
- Interested in affordable, practical, protein-rich foods.
- Prefer ready-to-eat or no-cook calorie-dense snacks when convenient.
- Need food suggestions that fit around gym sessions and work.

Foods mentioned in previous conversations

- Milk.
- Oats.
- Dry fruits and nuts.
- Peanuts.
- Chana.
- Sprouts.

- Soya curry.
- Besan and besan chilla.
- Chapatis and sabji.
- MyProtein shake after the gym.

I have discussed soya curry for office lunch, approximately 50 g of dry soya per serving in one conversation. A possible target of approximately 90 g dry soya per day was also considered, but it should not be treated as a confirmed daily intake.

I have considered using around 50 g of besan to prepare besan chilla. My dinner has often consisted of approximately five to six chapatis with sabji, according to an earlier discussion.

Nutrition planning preferences

When recommending food:

- Respect my vegetarian diet.
- Avoid making paneer or tofu the main protein sources.
- Consider convenient and affordable foods.
- Consider calorie intake as well as protein intake when discussing weight gain.
- Explain estimated protein and calories with clear assumptions.
- Distinguish dry food weight from cooked weight.
- Avoid assuming that previously discussed portions represent my current daily intake.

These details are for personalization, not a medical diagnosis or a verified nutrition log.

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8. Fitness and Daily Routine

In September 2026, I shared the following fitness context:

- Height: approximately 180 cm.
- Weight: approximately 64 kg.
- Gym experience: approximately one year.
- Goal: clean weight gain.
- Typical gym duration: approximately 1.5–2 hours per day.
- Sunday: usually a rest day.

I have also discussed aiming for approximately four hours of study on weekdays and two to four hours on Sundays.

My study topics include DSA, Java, Spring Boot, microservices, and DevOps.

When suggesting routines, account for the fact that I balance professional work, interview preparation, technical learning, and gym sessions. Ask me to confirm my current schedule if exact timings materially affect the recommendation.

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9. DSA and Interview Preparation

I am actively preparing for Java backend interviews, including data structures and algorithms, Spring Boot, microservices, AWS, and system-design fundamentals.

My problem-solving preferences

I want to improve the process I use before writing code. I have identified a tendency to start coding before thoroughly examining constraints, performing a dry run, identifying edge cases, writing pseudocode, and choosing an appropriate data structure.

A helpful problem-solving process for me is:

1. Understand the problem and constraints.
2. Work through a small example manually.
3. Identify edge cases.
4. Consider brute-force and optimized approaches.
5. Select the appropriate data structure or algorithm.
6. Write pseudocode.
7. Implement the solution in Java.
8. Prove correctness.
9. Analyze time and space complexity.
10. Test the solution with representative cases.

I prefer detailed dry runs, proofs, and explanations of why an algorithm works.

When I ask for help fixing my code, preserve my original approach whenever reasonably possible instead of replacing it entirely.

Topics discussed

- Dynamic programming.
- Longest increasing subsequence.
- Longest string chain.
- Minimum-score triangulation of a polygon.
- Sorted and rotated arrays.
- Sorting and digit-sum problems.

- Minimum swaps.
- Java streams and collections.
- Array and string algorithms.

I prefer explanations that clarify subtle details and do not rely on confusing memoization conventions without explaining them.

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10. Communication and Learning Style

Jarvis should adapt explanations to my technical level and the question being asked.

Preferred answer style

- Answer the direct question first.
- Explain the underlying reason.
- Use small, practical Java examples.
- Include output when it helps.
- Use tables for meaningful comparisons.
- Provide dry runs for algorithms.
- Explain edge cases and complexity when relevant.
- Be clear about assumptions and limitations.
- Avoid unnecessary repetition.
- Do not overwhelm a simple question with an excessively long explanation.

For implementation-heavy topics, include code that is easy to understand and test.

For interview preparation, challenge my reasoning with realistic follow-up questions and help me distinguish a correct answer from an incomplete one.

For writing and resume tasks, prioritize accuracy, natural language, and claims I can confidently defend.

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11. Kolkata, Durga Puja, and Creative Preferences

I previously shared New Town, Kolkata as a location context while planning Durga Puja visits.

I have shown interest in pandals across Central, North, and South Kolkata, including route planning, distances, and suitable visiting times.

Cultural and creative interests

Pandal themes discussed include:

- Uttam Kumar, known in Bengal as Mahanayak.
- A red dragon theme.
- A Bengali children's comic theme remembered as having a meaning similar to "nonsense."
- Creature designs resembling matchsticks.
- Maa Durga and Kali.
- Kolkata's nighttime festive lighting.

Caption preferences

When I request WhatsApp status captions or social media captions:

- Keep the caption short, often one sentence.
- Use natural language with good vibes.
- Avoid cheesy, overly dramatic, or excessively filmy wording.
- English captions with suitable emojis are welcome.
- Bengali or Hinglish options may be useful when requested.
- Connect the caption to the actual image, pandal theme, or occasion.

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12. Rules Jarvis Should Follow

These rules should guide responses generated from this profile.

1. **Be factual.** Do not invent personal details such as age, current salary, or unmentioned qualifications.
2. **Use relevant context.** Retrieve only the personal information needed to answer the question.
3. **Respect privacy.** Do not reveal secrets, credentials, API keys, passwords, or access tokens.
4. **Keep facts current.** Treat weight, diet portions, routines, tools, and goals as potentially outdated.
5. **Preserve professional accuracy.** Do not turn a learning topic into a claim of professional implementation.
6. **Explain uncertainty.** If retrieved information is incomplete or contradictory, state the limitation or ask a focused follow-up.
7. **Teach, not just answer.** For technical questions, explain the reasoning and use practical examples.
8. **Personalize appropriately.** Use my vegetarian food preferences for nutrition suggestions and my Java/Spring background for relevant technical explanations.
9. **Avoid unnecessary disclosure.** Personal information should be used only when relevant to the user's request.
10. **Do not treat this document as exhaustive.** It summarizes available conversation context and may not reflect every aspect of my life or current situation.

13. Additional Details (Fill In)

> This section is intentionally left for details that were **not** present in the source profile. Replace the placeholders with accurate information so Jarvis never has to guess. Anything left as "Not provided" should be treated as unknown.

Education and background

- **Highest qualification / college:** Not provided.
- **Graduation year:** Not provided.
- **Certifications:** Not provided.
- **Languages spoken:** Not provided (Bengali, Hindi, and English are used in caption preferences, but fluency levels are not confirmed).

Work details

- **Current role title / band at TCS:** Not provided.
- **Tech stack on the current project:** Not provided beyond what is in Section 4.
- **Target companies or roles:** Not provided (Forward Deployed Engineer roles and Full Stack Java AI roles are of interest).
- **Target timeline for job switch:** Not provided.

Projects and links

- **GitHub profile:** Not provided.
- **LinkedIn profile:** Not provided.
- **Jarvis repository / tech choices (vector DB, embedding model):** Not provided.

Personal preferences

- **Hobbies outside tech and fitness:** Not provided.
- **Gym split / training program:** Not provided.
- **Wake-up / sleep schedule:** Not provided.
- **Food allergies or intolerances:** Not provided.

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14. Suggested RAG Metadata

For ingestion into a vector database, this document can be split into separate documents by section.

Suggested metadata:

- `profile_owner`: `Nihal Singh`
- `document_type`: `personal_knowledge_base`
- `source`: `user_context`
- `last_updated`: `2026-10-23`

Suggested categories:

- `personal_profile`
- `career_goals`
- `technical_skills`
- `professional_experience`
- `easybuy_project`
- `spring_ai`
- `rag_architecture`
- `ai_tools`
- `fitness`
- `vegetarian_diet`
- `dsa_preferences`
- `interview_preparation`
- `devops_learning`
- `kolkata_interests`
- `response_preferences`
- `additional_details`

Freshness policy: Personal measurements, food quantities, daily routines, and current goals should be reconfirmed when precision matters.

End of personal knowledge base.